

Bell Nozzle Rocket Engine Calculations

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Nomenclature

C_f	Vacuum thrust coefficient
$\dot{m}$	Mass flow rate
M	Molar mass
Ma	Mach number
$mol\%$	Mole percentage
P	Pressure
R	Gas constant
T	Temperature
v	Velocity
ϵ	Nozzle area ratio
γ	Ratio of specific heats
ρ	Density

Subscripts

a	Ambient
c	Chamber
e	Exit
s	Stagnation
t	Throat

Introduction

Documentation of calculations for a bell nozzle rocket engine

1 Propellant

1.1 Propellant Stoichiometry

2 Exhaust Gasses

2.1 Ratio of Specific Heats of Multi-Compound Gas Mixtures [1]

$$\frac{1}{\gamma - 1} = \sum \frac{mol\%}{\gamma_i - 1}$$

2.2 Stagnation Pressure of Exhaust Gasses [2]

$$P_s = P_1 \left(1 + \left(\frac{\gamma - 1}{2} \right) Ma_1^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

2.3 Stagnation Temperature of Exhaust Gasses [3]

$$T_s = T_c \left(1 + \left(\frac{\gamma - 1}{2} \right) Ma_1^2 \right)$$

3 Combustion Chamber Geometry

4 Throat Geometry

5 Nozzle Geometry

[1]

Bibliography

References

- [1] O. Lanzi, "How do you calculate the heat capacity ratio for a multi-compound gas?." <https://chemistry.stackexchange.com/a/166796>, 2022.
- [2] H. Vasishta, "Stagnation pressure." <https://www.youtube.com/watch?v=t98s1A7fvm0>, 2016.
- [3] H. Vasishta, "Stagnation temperature." <https://www.youtube.com/watch?v=Le5dgW4lMWs>, 2018.